

Aya Spira

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MSc student in Computer Science at Tel Aviv University, with a BSc in Computer Science from the Technion (magna cum laude). Researching the security of LLMs and AI agents since 2024, with experience in academic and applied research.

EXPERIENCE

Graduate Researcher, Tel Aviv University Oct 2025 – present

Research on the security of AI agents, focusing on tool use, hallucinated resources, untrusted external inputs, and black-box evaluation of agent behavior.

Researcher, IBM Research Israel, Data Security and Privacy Jan 2024 – Dec 2025

- Built the data and prompt leakage evaluation metrics in IBM's [Unitxt](#)-based stack, used internally against the watsonx AI risk atlas.
- Built a working internal prototype of authorization-aware semantic search over [Milvus](#): ACLs with inheritance and deny rules, pre-filtered so a search only reaches permitted documents.
- Contributed to an [agent-based system for database access decisions](#).

R&D Intern, PTC Oct 2022 – Apr 2023

Developed features for a production 3D CAD system in C/C++, including components of its web layer in JavaScript and Three.js.

RESEARCH

Beware of Agentic Botnets: Scalable Untargeted Promptware Attacks via Adversarial HalluSquatting, First Author, arXiv preprint 2026

Studied hallucinated resource identifiers across LLMs and production coding agents, showing how predictable hallucinations can be exploited for AI supply-chain attacks. [Project page](#). Covered by [Ars Technica](#), [Fox News](#), [SecurityWeek](#), and [The Hacker News](#).

Visually Guided Object Insertion into Images, Bachelor's Project, Geometric Image Processing Lab, Technion2024

Built a dataset and trained a diffusion model for reference-guided object insertion, extending [Paint by Inpaint](#) (CVPR 2025). Finalist, Faculty Excellent Projects Competition, 2025.

EDUCATION

M.Sc. Computer Science, Tel Aviv University Oct 2025 – present

GPA 94. Advisors: Prof. Avishai Wool and Dr. Ben Nassi.

B.Sc. Computer Science, Technion – Israel Institute of Technology Mar 2021 – Dec 2024

Magna cum laude, GPA 92.5. Advanced coursework in Computer Vision and Cybersecurity.

Exchange semester at TUM, Summer 2023.

CS Faculty Student Council, 2022–2024: Head of Staffing; Deputy Head of Events.

SKILLS

Security — AI and agent security, adversarial attacks, AI and software supply-chain security, access control and authorization, cryptography theory.

ML & Agents — LLM evaluation, agent behavior and tool-use analysis, cross-model evaluation, RAG and vector databases, metric and dataset design, agentic coding workflows, diffusion models.

Engineering — Python, C/C++, JavaScript, PostgreSQL, PyTorch, Hugging Face, NumPy, Docker, Linux/Bash, Git, Milvus, FAISS, pandas.

Languages — Hebrew (native), English (fluent), German (A2).

PROGRAMS & AWARDS

[CMMRS 2026](#), Max Planck Institute for Security and Privacy, Saarbrücken.

2nd Place, [CS Hackathon: Doing Good 2023](#), Technion.